



Solutions Guide: Distro Research

Questions & Answers

1. Which distribution is most flexible and best suited for day-to-day and administrative tasks?
 - **Solution:** Ubuntu
2. Which distribution is built specifically for penetration testers?
 - **Solution:** Kali
3. Which distributions would you use to set up a web or data server?
 - **Solution:** Really, you could use any of them. However, Ubuntu Server and Fedora Server both have easy-to-configure web services. If you wanted to do things more manually, you could use Debian or CentOS.
4. What is the most widely used Linux desktop environment?
 - **Solution:** Ubuntu

B. Use Cases

- **Central Data Server** – This server stores sensitive human resources data, so it should use a secure distribution. Fedora and CentOS use SELinux by default, making them good choices.
- **Public Web Server** – This server needs to run quickly and handle large amounts of traffic. Many distributions could work, but Ubuntu and Fedora are among the best choices because they are well-supported.
- **IT Audit Workstation** – This workstation should use Kali Linux, because it is specifically designed for performing security assessments.
- **User Workstations** – This workstation should use Ubuntu, which comes with a GUI and basic productivity software, such as an email client, web browser, and text editor.

Bonus Questions & Answers

1. What is a "headless server"?
 - **Solution:** A command line-only server without a desktop environment. Does Ubuntu make a headless server variant? What about Fedora? CentOS?
 - **Solution:** Yes, all three of them do.
2. Which distribution is Ubuntu based on? What about Kali?
 - **Solution:** They are both based on Debian.
3. Which distribution is CentOS based on? What about Fedora?

- **Solution:** They are both based on Red Hat.
- 4. What is SELinux? Which distributions implement SELinux by default?
 - **Solution:** SELinux is a built-in file permissions security enhancement developed by the NSA. CentOS and Fedora have it implemented by default.
- 5. If you were deciding between versions of Ubuntu Server and wanted one that will remain stable over time, which version would you choose?
 - **Solution:** You would choose the "Long Term Support" or *LTS* version. The latest version is one that will have continual updates and changes. The LTS version will remain stable and only changes about once a year.
- 6. What are some security implications of using free and open-source software or forks of popular Linux distributions?
 - **Solution:** As demonstrated in the Mint OS article, *open source* means that anyone can contribute. This makes it somewhat easy for an attacker with programming skills to attack. Because of this, users must be vigilant about where they get their software.

Summary

Note the following takeaways from this activity:

- Most Linux distributions are forks of Debian or Red Hat.
- Ubuntu is the most common general-purpose Linux distribution, while Kali is designed specifically for security professionals.

With these distros, you can accomplish almost anything you need with Linux.